

NFR Template Card

Day 3 handout. The four-part shape every NFR follows, plus the five categories you must cover (at least one NFR each). The **Defense** is what separates a TPM-quality NFR from boilerplate — it's the part that's actually the work.

The four-part shape

```
### NFR – <short name>
**Category:** Performance / Security / Accessibility / Observability / Compliance

**Requirement:** <specific, measurable target>

**Defense:** <why this number, in plain language; what scenario justifies it>

**Verification:** <how we will test/observe this in production or pre-prod>
```

A good defense reads like this: *"p95 latency < 400ms because dispatchers tap this 40 times per shift; 1 second feels slow at that frequency, and 400ms is below the threshold of feeling sluggish."* That reasoning is the work.

The five categories — cover all five

CATEGORY	WHAT IT PINS DOWN
1. Performance	Latency targets (p50/p95/p99), throughput, resource caps, cold-start time-to-interactive
2. Security	AuthN / AuthZ, data classification (PII/PCI/internal/public), encryption in transit + at rest, audit logging
3. Accessibility	WCAG conformance level (2.1 AA is the floor), the 8 checks, feature-specific keyboard + screen-reader behavior
4. Observability	Which logs (events, levels, structured fields), which metrics (from your Tier Sheet), which traces, which alerts
5. Compliance	Data residency, retention windows, industry regimes (SOC 2, HIPAA, PCI), audit-trail requirements

Most PRDs need **6–10 NFRs total**.

Worked FieldPulse example

```
### NFR – Reconcile flow latency
**Category:** Performance
**Requirement:** Modal opens within 1 second (p95) on the median dispatcher device
                  (mid-tier Android, 2-year-old hardware) on a 4G LTE connection.
**Defense:** Dispatchers tap "Reconcile all" 30+ times per shift. At p95 = 1s,
               cumulative wait ≤ 30s/shift. At p95 = 3s, cumulative wait = 90s/shift,
               crossing the threshold dispatchers reported as "I'd rather use paper."
**Verification:** Synthetic transaction in Datadog from a fleet of mid-tier devices.
                  p95 reported daily; alert if 7-day trailing p95 > 1.2s.
```

The good-NFR self-check

- ☐ Specific — names a number, a screen, a scope, a regime.
- ☐ Measurable at runtime — not a wish.
- ☐ Defended — grounded in user behavior or a metric, not abstraction.
- ☐ Testable — the Verification line is real, not aspirational.
- ☐ Performance targets are at **p95 or p99**, never "average" (average lies in bimodal distributions).